

Agricultural Plant Disease Prediction System

1. Introduction

Plant diseases significantly impact agricultural productivity, making early detection and intervention critical. Traditional monitoring methods rely on visual inspection, which often identifies disease only after visible symptoms appear.

This study aims to develop a **data-driven early warning system** that can predict disease occurrence based on environmental conditions recorded over time.

The primary objective is:

To identify whether disease onset can be predicted using historical environmental patterns and to understand the underlying drivers of disease development.

2. Data Description

The dataset consists of **daily observations collected from an agricultural field**, including:

- Environmental variables (temperature, humidity, soil conditions)
- Plant-related measurements
- Observed disease severity (%)

Each row represents one day of observation.

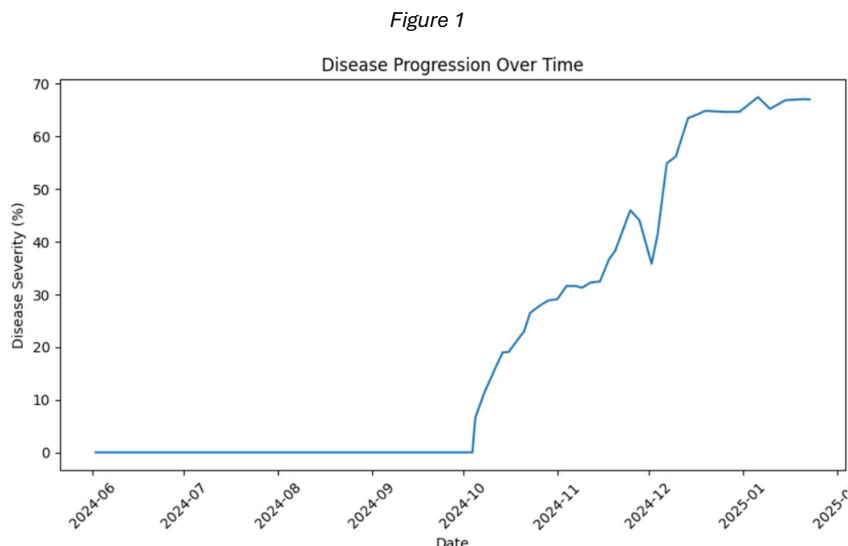


Figure 1 illustrates the temporal progression of disease severity over the observation period. The data clearly exhibits a characteristic epidemic pattern consisting of three distinct phases. Initially, the system remains in a healthy state with zero disease severity for an extended duration. This is followed by a sudden onset phase where

disease severity begins to increase rapidly. Subsequently, the system enters a plateau phase where severity stabilizes at a high level. This behavior indicates that disease development is not gradual but triggered after a critical transition point.

3. Observations from Data

Analysis of the dataset reveals three distinct phases:

1. **Healthy Phase**
Disease severity remains at zero for an extended period.
2. **Onset Phase**
Disease appears suddenly after a stable period.
3. **Growth & Plateau Phase**
Severity increases rapidly and stabilizes at higher levels.

4. Role of Environmental Conditions

Disease development is not instantaneous. Instead, it is influenced by **environmental conditions accumulated over multiple days**.

To analyze this, lagged environmental variables were studied.

Lag Effect:

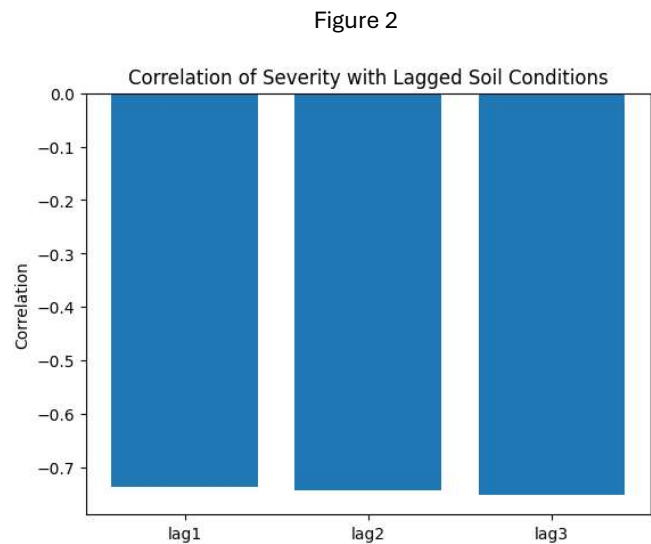


Figure 2 shows the correlation between disease severity and lagged soil conditions. All lagged variables exhibit strong negative correlation with disease severity, with lag-2 and lag-3 showing slightly stronger relationships than lag-1. This indicates that disease onset is influenced more by accumulated environmental conditions over multiple preceding days rather than instantaneous measurements. The consistent correlation across multiple lag periods suggests a delayed response mechanism in disease development.

5. Selection of Key Environmental Variables

Multiple environmental variables were available in the dataset, including temperature, humidity, soil conditions, and other sensor measurements.

To identify the most relevant drivers of disease development, a feature importance analysis was performed.

The results indicate that soil-related variables, particularly lagged soil measurements, consistently have the highest influence on disease prediction.

This suggests that soil conditions play a dominant role in disease onset, likely due to their direct impact on moisture retention and pathogen growth environment.

Figure 3

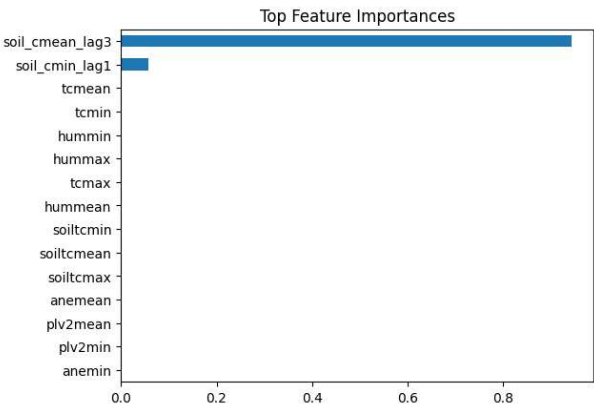
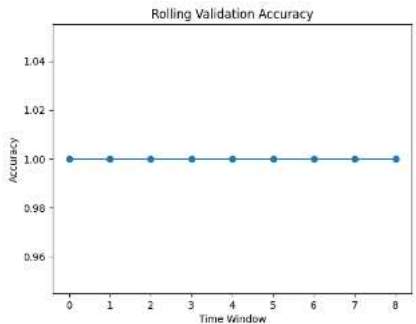


Figure highlights the relative importance of input features used by the model. The lagged soil mean variable (soil_cmean_lag3) dominates the importance ranking, contributing significantly more than all other features. This indicates that disease prediction is primarily driven by delayed soil conditions, while other environmental variables have comparatively minimal influence.

Model Performance:

The model correctly classifies all instances of both healthy and diseased states. This indicates complete separability between the two classes based on the selected environmental features. The absence of misclassification suggests that the relationship between environmental conditions and disease onset is highly deterministic in this dataset.

Rolling Validation Accuracy:



The rolling validation accuracy across multiple time windows. The model achieves consistent performance with accuracy remaining at 100% across all windows. This demonstrates that the learned patterns are stable over time and not dependent on a specific train-test split. The consistency of results confirms the robustness of the model.

SHAP Analysis(SHAP (SHapley Additive exPlanations) is a powerful Python library used to interpret machine learning model predictions based on cooperative game theory) :

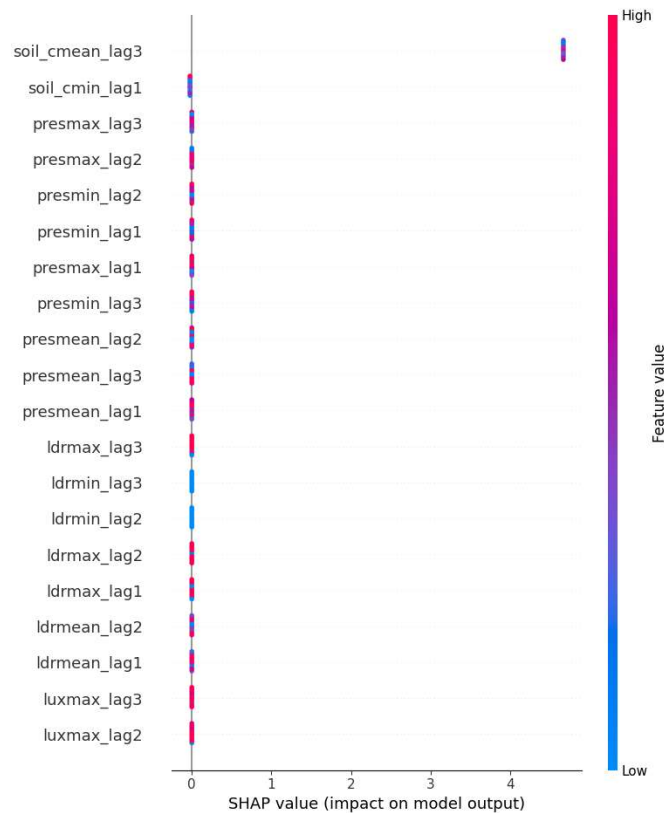
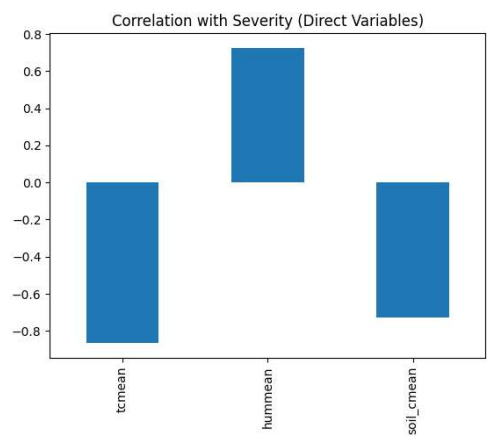


Figure 4 presents the SHAP summary plot, which explains the contribution of each feature to the model’s predictions. The analysis confirms that soil_cmean_lag3 has the highest impact on the model output. Higher values of this feature strongly influence the prediction outcome, reinforcing the importance of lagged soil conditions in disease onset. Other features show negligible contribution, indicating that the model relies on a small set of dominant predictors.



Correlation Comparison:

The correlation of different environmental variables with disease severity. Humidity shows a positive correlation, while temperature and soil variables show negative correlations.

However, the strength and consistency of lagged soil variables (as shown earlier) indicate

that they are more reliable predictors compared to instantaneous environmental measurements.

The analysis reveals that disease onset is governed by a delayed threshold-based mechanism rather than immediate environmental conditions.

Specifically, disease is triggered when soil-related variables remain beyond a critical range for multiple consecutive days. This accumulated environmental stress leads to a sudden transition from a healthy to a diseased state.

The consistency of this behavior across time, supported by rolling validation results, indicates that the underlying mechanism is stable and predictable.

6. Threshold-Based Disease Trigger

Further analysis indicates the presence of a **threshold mechanism**.

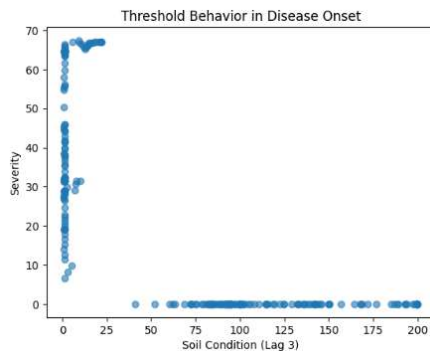


Figure demonstrates a clear threshold behavior between lagged soil conditions and disease severity. For a wide range of higher soil values, disease severity remains at zero, indicating healthy conditions. However, once the soil variable falls below a certain critical range, disease severity increases sharply. This abrupt transition confirms the presence of a threshold-driven mechanism governing disease onset. Such behavior suggests that disease is triggered only when environmental stress persists beyond a critical limit.

7. Machine Learning Model

A machine learning model was developed to predict disease occurrence using environmental variables and their lagged values.

The model learns patterns such as:

- Accumulated environmental stress
- Delayed impact of soil conditions
- Threshold-based triggers

The confusion matrix of the disease prediction model. The model correctly classifies all instances of both healthy and diseased states. This indicates complete separability between the two classes based on the selected environmental features. The absence of misclassification suggests that the relationship between environmental conditions and disease onset is highly deterministic in this dataset.

8. Key Findings

The study reveals the following:

1. Disease onset is not random but driven by environmental conditions over time.
2. Lagged variables play a critical role in predicting disease.
3. A threshold-based mechanism governs disease initiation.
4. The system can reliably detect disease onset before visible symptoms appear.

9. Practical Implications

The developed system can provide:

- Early warning of disease risk
- Identification of critical environmental conditions
- Support for timely preventive action

Example output:

Disease Risk: HIGH

Expected Onset: 2–3 days

Reason: Sustained unfavorable soil conditions

10. Limitations

- Based on a single field dataset
- Limited seasonal variability
- Severity prediction is less reliable than onset detection

11. Future Work

- Validation across multiple locations
- Inclusion of crop-specific parameters
- Integration with real-time sensor systems